

Last Time

T-tests of One Mean

Estimating σ^2 , revisited

Behrens-Fisher Problem

Today

Two Independent Samples
Sampling Distribution of $\bar{X}_1 - \bar{X}_2$

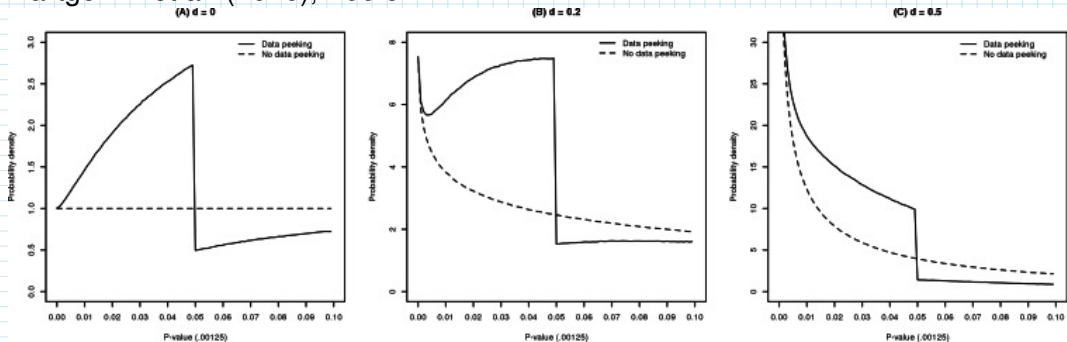
Behrens-Fisher Problem

Dependent Samples

Advantages to dependent sample designs

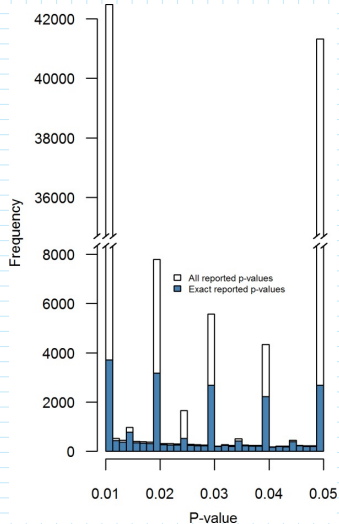
P-values in Published Research

Hartgerink et al. (2016), *PeerJ*



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GSS Example

Can money buy happiness? Look at the income and happiness variables in the GSS dataset.

- ▶ Collapse the happiness rating (1=very happy, 2=pretty happy, 3=not too happy) into happy vs. not happy (two groups).
- ▶ If money buys some happiness, the income for the happy group should be higher than the income for the not happy group.

```
data <- read.csv('GSS_1980.csv')  
  
# Collapse HAPPY variable and rename income  
happy <- data$HAPPY < 3  
money <- data$REALRINC
```

Income Statistics

	Happy	Unhappy
N	770	103
$\bar{X}$	\$23,849.19	\$17,924.87
s	\$24,780.02	\$17,665.58

Sampling Distribution of $\bar{X}_1 - \bar{X}_2$

We know that $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/N)$.

What is the sampling distribution of $\bar{X}_1 - \bar{X}_2$?

Assume:

- ▶ All $\bar{X}_i \sim \mathcal{N}(\mu_i, \sigma^2/N_i)$: different groups of different sizes have different means, but all groups have the same population variance σ^2 .
- ▶ The groups are independent: different people are in each group and they are not connected to each other in any way.

Also: $X \sim \mathcal{N}(\mu, \sigma^2)$

Distribution of $\bar{X}_1$ and $\bar{X}_2$

Because both $\bar{X}_1$ and $\bar{X}_2$ are normally distributed, $\bar{X}_1 - \bar{X}_2$ is normally distributed. Therefore,

$$\bar{X}_1 - \bar{X}_2 \sim \mathcal{N}(\mu_1 - \mu_2, \sigma_1^2/N_1 + \sigma_2^2/N_2)$$

- ▶ $H_0 : \mu_1 = \mu_2 \rightarrow H_0 : \mu_1 - \mu_2 = 0$
- ▶ What to do about $\sigma_1^2/N_1 + \sigma_2^2/N_2$?
- ▶ (Note that if $\sigma_1^2 = \sigma_2^2 = \sigma^2$, then $\sigma^2/N_1 + \sigma^2/N_2 = \sigma^2(1/N_1 + 1/N_2)$.)

Estimating the Standard Error

1. $\frac{\sigma^2}{N_1} + \frac{\sigma^2}{N_2} \Rightarrow \frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}$
2. Pool the variance: devise a single estimate of the common σ^2 using both samples

Pooling the Variance

For two samples X and Y ,

$$s_1^2 = \frac{\sum_{i=1}^{N_1} (X_i - \bar{X})^2}{N_1 - 1} \quad s_2^2 = \frac{\sum_{i=1}^{N_2} (Y_i - \bar{Y})^2}{N_2 - 1}$$

Call the numerators the sums of squares for X and Y : SSX and SSY .
Then

$$s_P^2 = \frac{SSX + SSY}{(N_1 - 1) + (N_2 - 1)}$$

Pooling the Variance

Another way to think about it:

$$s_P^2 = \frac{(N_1 - 1)s_1^2 + (N_2 - 1)s_2^2}{(N_1 - 1) + (N_2 - 1)}$$

A weighted average of the two sample variances.

$$\hat{\sigma}^2 = s_P^2$$

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GSS Example

Do happy people have higher incomes than unhappy people?

$$H_0 : \quad \mu_H = \mu_U \text{ (or } \mu_H - \mu_U = 0 \text{)}$$

$$H_1 : \quad \mu_H > \mu_U \text{ (or } \mu_H - \mu_U > 0 \text{)}$$

- ▶ Assume $\sigma_H^2 = \sigma_U^2$.
- ▶ **Assume normality of the income measurements.**
- ▶ Choose $\alpha = 0.05$.
- ▶ Estimate σ^2 with s_p^2 .
- ▶ Compute the t -statistic for the mean difference.
- ▶ Compute the p -value of the t -statistic.

Independent Samples t -Test

$$\bar{X}_H - \bar{X}_U = \$5,924.32$$

$$\begin{aligned}s_P^2 &= \frac{(770 - 1)(614,049,163) + (103 - 1)(312,072,780)}{(770 - 1) + (103 - 1)} \\ &= 24,055.89\end{aligned}$$

$$\hat{\sigma}_{\bar{X}_H - \bar{X}_U} = s_P \sqrt{\frac{1}{N_H} + \frac{1}{N_U}} = 2523.86$$

$$T = \frac{5924.32 - 0}{2523.86} = 2.3473$$

$$p = 0.0096$$

Independent Samples *t*-Test

```
> t.test(money~happy, alternative="less", var.equal=TRUE)
```

Two Sample *t*-test

data: money by happy

$t = -2.3473$, $df = 871$, $p\text{-value} = 0.009566$

alternative hypothesis: true difference in means between group
FALSE and group TRUE is less than 0

95 percent confidence interval:

-Inf -1768.53

sample estimates:

mean in group FALSE	mean in group TRUE
17924.87	23849.19

What if $\sigma_1^2 \neq \sigma_2^2$?

- ▶ Homoskedasticity: $\sigma_1^2 = \sigma_2^2$
- ▶ Heteroskedasticity: $\sigma_1^2 \neq \sigma_2^2$

Behrens-Fisher Problem: $\sigma_1^2 \neq \sigma_2^2$

$$T = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}}$$

Satterthwaite Correction

$$T' = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}} \sim t(df')$$

where

$$df' = \frac{\left(\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}\right)^2}{\frac{\left(\frac{s_1^2}{N_1}\right)^2}{N_1 - 1} + \frac{\left(\frac{s_2^2}{N_2}\right)^2}{N_2 - 1}}$$

Welch's Two-Sample t -Test

$$df' = 161.278$$

$$\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}} = 1956.348$$

$$T' = \frac{5924.32 - 0}{1956.348} = 3.028$$

$$p = Pr(T > T' \mid df = df') = 0.0014$$

Welch's Independent Samples t -Test

```
> t.test(money~happy, alternative="less", var.equal=FALSE)
```

Welch Two Sample t -test

data: money by happy

$t = -3.0283$, $df = 161.28$, $p\text{-value} = 0.001432$

alternative hypothesis: true difference in means between group
FALSE and group TRUE is less than 0

95 percent confidence interval:

-Inf -2687.825

sample estimates:

mean in group FALSE	mean in group TRUE
17924.87	23849.19

Which to Use

Remember assumptions:

- ▶ Samples are drawn from normal populations
- ▶ Homoskedasticity: $\sigma_1^2 = \sigma_2^2$
- ▶ Equal sample sizes (oops)
- ▶ T-test is robust to moderate violations of normality and homoskedasticity; the Type I error rate doesn't get much bigger than the α level.
 - ▶ Sample sizes need to be comparable
 - ▶ Very unequal sample sizes should use Welch's
- ▶ R uses Welch's test by default, because it is now standard practice.
- ▶ If you're worried, you can perform one of a number of tests for equality of variances, but making this routine can inflate Type I error rates (Hayes & Cai, 2007).

Tests for Equality of Variances

- ▶ Levene's test (`DescTools::LeveneTest(money~happy)`)
- ▶ Fligner-Killeen test (`stats::fligner.test(money~happy)`)
- ▶ F ratio (`stats::var.test(money~happy)`)
- ▶ Bartlett test (`stats::bartlett.test(money~happy)`)

Dependent Samples

Participant	Treatment 1	Treatment 2
1	X_1	Y_1
2	X_2	Y_2
3	X_3	Y_3
...		
N	X_N	Y_N
Mean	$\bar{X}$	$\bar{Y}$

Alzheimer's Diagnostic

Finger tapping speed can be an indication of a number of neurological disorders. Christianson and Leathem (2004) developed a computerized diagnostic test that, for healthy individuals, has a mean of $\mu = 59$ and a standard deviation of $\sigma = 7$.

A sample of $N = 10$ Alzheimer's patients takes the test, and we observe their mean $\bar{X} = 48.90$ and standard deviation $s = 13.36$.

One-sample t -Test

```
speed <- c(36, 46, 76, 58, 49, 35, 46, 51, 60, 32)
t.test(speed, mu=59, alternative='less')
```


New Research Question

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A sample of $N = 10$ Alzheimer's patients takes the test, and we observe their mean $\bar{X} = 48.90$ and standard deviation $s_X = 13.36$. They are then placed on a new drug regimen, and after three months on the new drug they are retested. The second mean $\bar{Y} = 51.20$ and $s_Y = 12.28$.

Difference Scores

Patient	1	2	3	4	5	6	7	8	9	10
Before Treatment	36	46	76	58	49	35	46	51	60	32
After Treatment	42	46	75	65	55	33	49	53	54	40

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Before Treatment	36	46	76	58	49	35	46	51	60	32
After Treatment	42	46	75	65	55	33	49	53	54	40
Difference	-6	0	1	-7	-6	2	-3	-2	6	-8

Before	After
$\bar{X} = 48.90$	$\bar{Y} = 51.20$

$$\bar{D} = -2.3$$

$$s_D = 4.55$$

$$N = 10$$

Repeated or Dependent Measures t -test

Before	After
$\bar{X} = 48.90$	$\bar{Y} = 51.20$
$\bar{D} = -2.3$	
$s_D = 4.55$	
$N = 10$	

In R

```
> before <- c(36,46,76,58,49,35,46,51,60,32)
> after <- c(42,46,75,65,55,33,49,53,54,40)
> t.test(x=speed,y=before,mu=0,paired=TRUE,alternative='less')
```

Paired t-test

```
data: before and after
t = -1.5995, df = 9, p-value = 0.07209
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
    -Inf 0.3359742
sample estimates:
mean difference
    -2.3
```

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$$\sigma_{\bar{X}-\bar{Y}} = \sqrt{\sigma_{\bar{X}}^2 + \sigma_{\bar{Y}}^2 - 2\rho\sigma_{\bar{X}}\sigma_{\bar{Y}}}$$

If $\rho > 0$, the standard error is reduced.

- Typically, in matched samples or repeated measures, $\rho > 0$

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Alzheimer's disease *progresses*.

- ▶ Two groups: one group gets the drug first and is then taken off the regimen for retest, and the second group is placed on the regimen after the first test.
- ▶ While the drug gets time to have an effect, the disease gets worse.
- ▶ Some questions require controlling for order of treatments

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These disadvantages can usually be overcome with good experimental design.